ECE253/CSE208 Introduction to Information Theory

Lecture 4: Mutual Information

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- Chap 2 of *Elements of Information Theory (2nd Edition)* by Thomas Cover & Joy Thomas

Mutual Information (MI)

Definition (Mutual Information)

Consider two random variables X and Y. The mutual information $I(X;Y)$ is the relative entropy between the joint distribution $p(x,y)$ and the product distribution $p(x)p(y)$:

$$\begin{aligned} I(X;Y) &= D_{\mathrm{KL}}\Big(p(x,y) \,||\, p(x)p(y)\Big) \\ &= \mathbb{E}_{p(x,y)}\left(\log \frac{p(X,Y)}{p(X)p(Y)}\right) \\ &= \sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log\frac{p(x,y)}{p(x)p(y)} \end{aligned}$$

MI Is Uncertainty Reduction

Let us rewrite $I(X;Y)$ as:

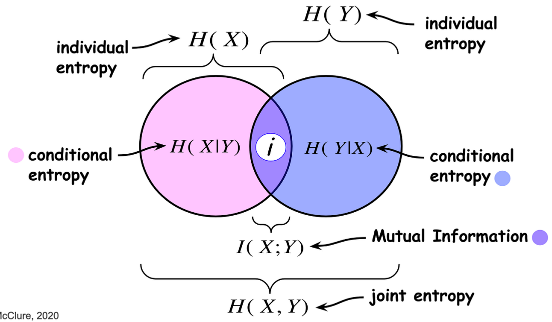
$$
\begin{aligned}
I(X;Y) &= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x,y) \log \frac{p(x|y)}{p(x)} \\
&= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x,y) \log p(x|y) - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x,y) \log p(x) \\
&= H(X) - H(X|Y)
\end{aligned}
$$

Insight:

$I(X;Y)$ is the reduction in the uncertainty of X due to the knowledge of Y. $I(X;Y) = I(Y;X) = H(X) - H(X|Y) = H(Y) - H(Y|X)$, which means that X says as much about Y as Y says about X.

MI Properties

- $I(X;X) = H(X) - H(X|X) = H(X)$.
- $I(X;Y) = H(Y) - H(Y|X) = H(X) + H(Y) - H(X,Y)$.
- Variation of information: $VI(X;Y) = H(X) + H(Y) - 2I(X,Y) = 2H(X,Y) - H(X) - H(Y) = H(X|Y) + H(Y|X)$.



Sean McClure, 2020

Mutual Information as a Measure of Dependence

- **Bounds**

$$0 \leq I(X;Y) \leq \min\{H(X), H(Y)\}$$

- **Independence characterization**

$$I(X;Y) = 0 \iff p(x,y) = p(x)p(y) \iff X \perp Y$$

- **Interpretation**
  - Measures *all* statistical dependence (not only linear)
  - Zero MI means knowing Y gives no information about X
  - Positive MI means X and Y share information
- **Contrast with correlation**
  - Zero correlation $\not\Rightarrow$ independence
  - Zero mutual information $\iff$ independence

Mutual information is the gold-standard dependence measure.

Conditional Mutual Information

Definition (Conditional mutual information)

The conditional mutual information of RVs X and Y given Z is defined by:

$$I(X;Y|Z) = H(X|Z) - H(X|Y,Z) = \mathbb{E}_{p(x,y,z)}\left(\log \frac{p(X,Y|Z)}{p(X|Z)p(Y|Z)}\right).$$

Interpretation

- Information shared by X and Y *after accounting for* Z.
- Reduction in uncertainty of X due to Y, given Z.
- $I(X;Y \mid Z) = 0 \iff X \perp Y \mid Z \quad \leftarrow \quad$ Given Z, X and Y are independent.

Conditioning removes information already explained by Z.

Example: Conditional MI

Setup: Consider $Z \sim \mathrm{Bern}(0.5)$; $\quad X = Z, \quad Y = Z.$

Unconditional MI

$$I(X;Y) = H(X) = 1 \text{ bit}$$

Conditional MI

$$H(X|Z) = 0, \quad H(X|Y,Z) = 0 \quad \Rightarrow \quad I(X;Y|Z) = 0$$

- X and Y are dependent, but conditionally independent given Z.
- All dependence between X and Y is explained by Z.

Chain Rule of Mutual Information

Theorem (Chain rule for mutual information)

$I(X_1, X_2, ..., X_n; Y) = \sum_{i=1}^{n} I(X_i; Y | X_{i-1}, X_{i-2}, ..., X_1)$.

Proof.

$$
\begin{aligned}
I(X_1, X_2, ..., X_n; Y) &= H(X_1, X_2, ..., X_n) - H(X_1, X_2, ..., X_n | Y) \\
&= \sum_{i=1}^{n} H(X_i | X_{i-1}, X_{i-2}, ..., X_1) - \sum_{i=1}^{n} H(X_i | X_{i-1}, X_{i-2}, ..., X_1, Y) \\
&= \sum_{i=1}^{n} I(X_i; Y | X_{i-1}, X_{i-2}, ..., X_1)
\end{aligned}
$$

$\square$

Mutual Information: Invariance and Additivity

- **Invariance to bijections:** If f and g are invertible transformations, then

$$I(X;Y) = I\big(f(X); g(Y)\big).$$

  Mutual information depends only on the *statistical dependence*, not on the particular parameterization.

- **Contrast with entropy:** Unlike (differential) entropy, mutual information is invariant under reparameterization.

- **Additivity for independent pairs:** If (X_1, Y_1) and (X_2, Y_2) are independent,

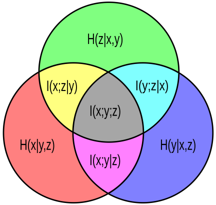
$$I\big((X_1, X_2); (Y_1, Y_2)\big) = I(X_1; Y_1) + I(X_2; Y_2).$$

- **Interpretation:** Independent information sources contribute additively to the total mutual information.

Interaction Information (1/2)

Definition (via Conditional Mutual Information)

$$I(X;Y;Z) \triangleq I(X;Y) - I(X;Y \mid Z)$$



Interpretation

- Measures how a third variable Z modifies the relationship between X and Y
- Captures higher-order (three-way) statistical interaction

Interaction Information (2/2)

Equivalent Entropy Expansion (Symmetric Form)

$$I(X;Y;Z) = \underbrace{H(X) + H(Y) + H(Z)}_{\text{individual entropies}} - \underbrace{\big(H(X,Y) + H(X,Z) + H(Y,Z)\big)}_{\text{pairwise joint entropies}} + \underbrace{H(X,Y,Z)}_{\text{full joint entropy}}.$$

Equivalent Probability (Log-Ratio) Form

$$I(X;Y;Z) = \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y)\,p(x,z)\,p(y,z)}{p(x,y,z)\,p(x)\,p(y)\,p(z)}.$$

- Fully symmetric in X, Y, Z.
- Can be positive, zero, or negative.

Sign and Interpretation of Interaction Information

Case 1: $I(X;Y;Z) > 0$ **(Redundancy)**

- Conditioning on Z reduces the dependence between X and Y
- Z explains part of the correlation between X and Y
- Information shared by X and Y overlaps with information in Z
- Example: $Z \sim \text{Bern}(0.5), \quad X = Z,\ Y = Z \implies I(X;Y) > 0,\ I(X;Y \mid Z) = 0.$

Case 2: $I(X;Y;Z) < 0$ **(Synergy)**

- Conditioning on Z reveals dependence not visible marginally
- X and Y become informative only when considered jointly
- Example: $X \perp Y,\ Z = X \oplus Y \implies I(X;Y) = 0,\ I(X;Y \mid Z) > 0.$

Case 3: $I(X;Y;Z) = 0$

- No higher-order interaction beyond pairwise dependencies

MI Application 1: Feature Selection

Problem

- Given features $X_1, \ldots, X_n$ and label Y
- Select features that are most informative about Y

MI-based relevance: *Good features reduce uncertainty about the label.*

$$\text{Score}(X_i) = I(X_i; Y)$$

Why mutual information?

- Captures nonlinear dependence
- Model-agnostic (any supervised learner): Linear models, tree-based models, neural networks, etc.

Redundancy control

$$I(X_i; Y \mid X_j) \approx 0 \;\Rightarrow\; X_i \text{ adds little new information}$$

MI Application 2: Decision Tree

A tree structure: each *internal node* denotes a test on an attribute, each *branch* represents an outcome of the test, and each *leaf* represents a class (or class distribution).

Q: Which attribute to choose for splitting?

A: Choose the most relevant attribute for classification.

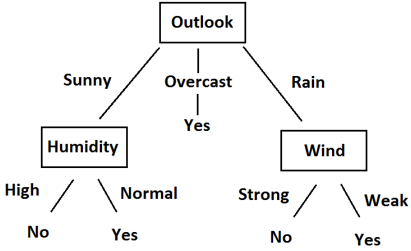


Figure: A decision tree to decide if we will play tennis.

Information Gain (IG)

Outlook	Temperature	Humidity	Windy	Play
Sunny	Hot	High	False	***No***
Sunny	Hot	High	True	***No***
Overcast	Hot	High	False	***Yes***
Rainy	Mild	High	False	***Yes***
Rainy	Cool	Normal	False	***Yes***
Rainy	Cool	Normal	True	***No***
Overcast	Cool	Normal	True	***Yes***
Sunny	Mild	High	False	***No***
Sunny	Cool	Normal	False	***Yes***
Rainy	Mild	Normal	False	***Yes***
Sunny	Mild	Normal	True	***Yes***
Overcast	Mild	High	True	***Yes***
Overcast	Hot	Normal	False	***Yes***
Rainy	Mild	High	True	***No***

Figure: 5 no-play and 9 play. Let $Y \sim \text{Bern}(\frac{9}{14})$ be the distribution of the play-or-not decision. Without splitting: $H(Y) = 0.94$ bits.

Outlook	Temperature	Humidity	Windy	Play
Sunny	Hot	**High**	False	***No***
Sunny	Hot	**High**	True	***No***
Overcast	Hot	**High**	False	***Yes***
Rainy	Mild	**High**	False	***Yes***
Rainy	Cool	**Normal**	False	***Yes***
Rainy	Cool	**Normal**	True	***No***
Overcast	Cool	**Normal**	True	***Yes***
Sunny	Mild	**High**	False	***No***
Sunny	Cool	**Normal**	False	***Yes***
Rainy	Mild	**Normal**	False	***Yes***
Sunny	Mild	**Normal**	True	***Yes***
Overcast	Mild	**High**	True	***Yes***
Overcast	Hot	**Normal**	False	***Yes***
Rainy	Mild	**High**	True	***No***

Figure: The information gain by splitting the attribute $X =$ Humidity is: $IG(\text{Humidity}) = H(Y) - H(Y|X)$.

Information Gain of Each Feature

$$
\begin{aligned}
H(Y|X) &= \Pr(X = \text{high}) \times H(Y|X = \text{high}) + \Pr(X = \text{normal}) \times H(Y|X = \text{normal}) \\
&= \frac{7}{14} \times H\left(\frac{3}{7}, \frac{4}{7}\right) + \frac{7}{14} \times H\left(\frac{1}{7}, \frac{6}{7}\right) \\
&= 0.152
\end{aligned}
$$

Similarly, we can calculate

$$
IG(\text{outlook}) = 0.247, IG(\text{temperature}) = 0.029, IG(\text{windy}) = 0.048
$$

Thus, **outlook** has the highest IG. We select it for the initial split to start building the decision tree.

Split via the Highest Information Gain

Now search for the best split at the next level:

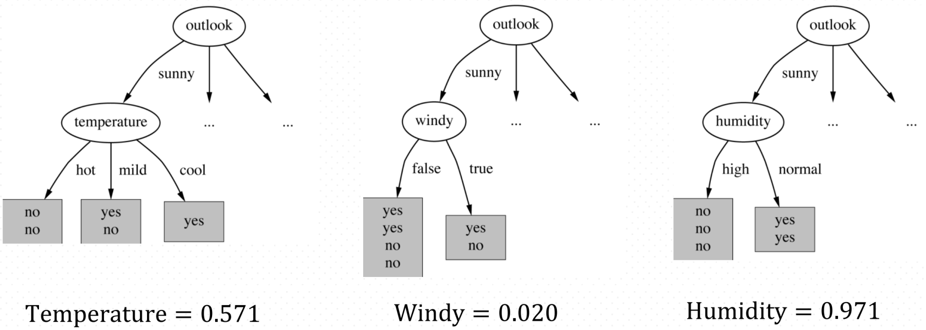


Figure: Humidity has the highest info gain at the second level when outlook = sunny.

The Built Decision Tree

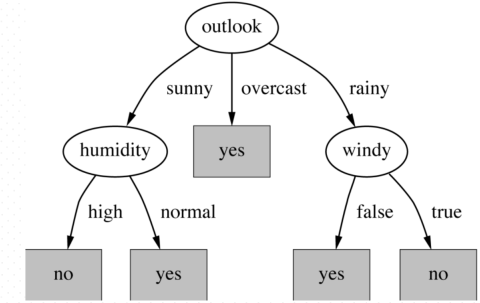


Figure: The built decision tree.

- Not all leaves need to be pure.
- Sometimes similar (even identical) instances have different classes.
- Splitting stops when data cannot be split any further.

MI Application 3: Deep learning

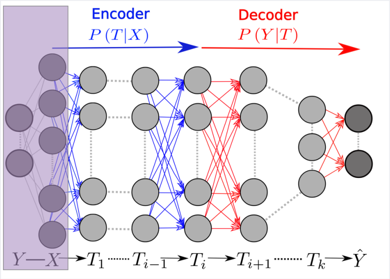


Figure: The DNN layers form a Markov chain of successive internal representations of the input layer X. Any representation of the input, T, is defined through an encoder, $P(T|X)$, and a decoder $P(\hat{Y}|T)$, and can be quantified by its information plane coordinates: $I_X = I(X;T)$ and $I_Y = I(T;Y)$. [1]

[1] Shwartz-Ziv and Tishby, "Opening the Black Box of Deep Neural Networks via Information," 2017.

Information Plane

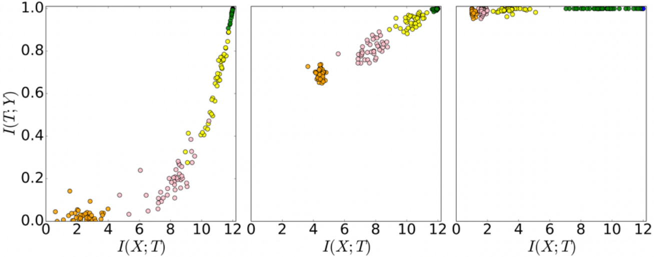


Figure: Snapshots of layer representations from multiple randomized networks during SGD training, shown in the information plane at initialization, intermediate, and late training stages. Mutual information is invariant to invertible reparameterizations, making it suitable for studying learned representations.

Key question: How does information flow and transform through layers during training?

Two Phases of Learning Revealed by Mutual Information

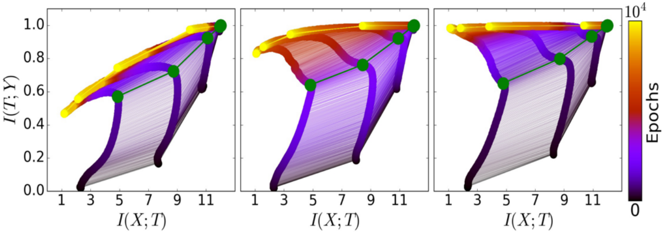


Figure: The evolution of the layers with the training epochs in the information plane, for different training samples. Left, middle, and right correspond to 5%, 45%, and 85% of the data, respectively.

- Training has *two distinct phases* in the Information Plane:
  1. **Fitting phase (EMR: empirical risk minimization)** – Rapid increase in $I(T;Y)$; Network learns to predict labels.
  2. **Compression phase** – $I(X;T)$ decreases while $I(T;Y)$ remains high; Representations discard input information irrelevant to Y.

Information Bottleneck (IB) Bound

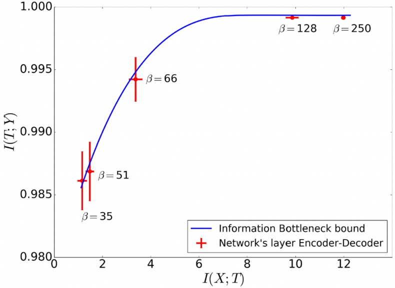


Figure: The optimal representations lie on the IB curve, which maximally compress the input X, for a given MI on the desired output Y.

- The IB principle formalizes representation learning:

$$\min_{p(t|x),\, p(y|t),\, p(t)} \{I(X;T) - \beta I(T;Y)\}$$

Information Bottleneck View of Deep Learning

- Compression arises naturally due to stochasticity in SGD. Most training time is spent in the compression phase.
- Empirically, trained DNN layers approach the **IB optimal tradeoff curve**.
- Deeper layers correspond to stronger compression (larger effective β).
- Interpretation:
  - Networks learn approximate **minimal sufficient representations**.
  - Generalization emerges from information compression rather than explicit regularization.

Key message: Deep learning works by compressing information while preserving relevance.

Mutual Information: Big Picture

- **Entropy** $H(X)$: uncertainty of a random variable
- **Conditional entropy** $H(X|Y)$: remaining uncertainty after observing Y
- **Mutual information**

$$I(X;Y) = H(X) - H(X|Y)$$

  measures the *reduction in uncertainty* of X due to Y

- **Equivalent viewpoints**
  - Dependence measure: $I(X;Y) = 0 \iff X \perp Y$
  - Divergence: $I(X;Y) = D_{\mathrm{KL}}(p(x,y) \,\|\, p(x)p(y))$
  - Coding: bits saved by joint encoding
- **Applications:** 1) Feature relevance and selection; 2) info gain in decision trees; and 3) info flows and bottleneck in neural networks

Mutual information is the fundamental measure of statistical dependence.